

COURSE TITLE : QUANTITY SURVEYING
COURSE CODE : 6125
COURSE CATEGORY : E
PERIODS PER WEEK : 4
SEMESTER : 6
PERIODS PER SEMESTER : 60
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
I	Introduction to Quantity Surveying	3
	Principles of Measurements	7
	Specification and Bill of material;s	6
	Test	1
II	Analysis of Price	7
	Tendering and Contracting	6
	Test	1
III	Laws of Surveyors	7
	Valuation of land and Building	6
	Test	1
IV	Project Management	6
	Computer application in quantity surveying	4
	Introduction to Standards	4
	Test	1
	Total	60

Course Outcome :

Module	G.O	Student will be able to:
1	1	Know Quantity Surveying
	2	Understand principles of Measurements and Description
2	1	Comprehend analysis of Price
	2	Understand Tendering and Contracting
3	1	Understand Laws of Surveyors
4	1	Understand Project management

Specific outcome

MODULE-I

1.1.0 Know Quantity Surveying

1.1.1 State Historical Development of QS

- 1.1.2 Describe the Functions of Quantity Surveyor
- 1.1.3 State Document
- 1.1.4 Explain Method of Computation of Measurement
- 1.1.5 State basic concept in QS

1.2.0 Understand principles of Measurements and Description

- 1.2.1 Explain Fundamentals in Engg. Drawing
- 1.2.2 Describe Bill of Quantities
- 1.2.3 Explain Preparation of Bill of Quantity
- 1.2.4 Explain Quantitative Inspection
- 1.2.5 Explain Preparation of Billing sheets
- 1.2.6

MODULE II

2.1.0 Comprehend analysis of Price

- 2.1.1 Define Cost
- 2.1.2 List Various Cost
- 2.1.3 Explain Standard schedule of rates
- 2.1.4 Describe Costing
- 2.1.5 Explain Factors affecting analysis of cost
- 2.1.6 Describe Economics of Hiring and owing

2.2.0 Understand Tendering and Contracting

- 2.2.1 State tender
- 2.2.2 List Different types of Tender
- 2.2.3 Explain Different Types of Tender
- 2.2.4 Describe Contract Conditions
- 2.2.5 Explain Contract Payment and Close out

MODULE III

3.1.0 Understand Laws of Surveyors

- 3.1.1 Comprehend Valuation of Land and Buildings
- 3.1.2 List Type of Property
- 3.1.3 State the Income from Property
- 3.1.4 List different Value

- 3.1.5 Explain Value
- 3.1.6 Describe Method of tax analysis

MODULE IV

4.1.0 Understand Project management

- 4.1.1 State the objectives of Project Management
- 4.1.2 Describe the Functions of Project Management
- 4.1.3 Describe Project Planning and Control
- 4.1.4 Explain Network Diagramme
- 4.1.5 Understand Computer Application in QS
- 4.1.6 State Quantity Estimation
- 4.1.7 Explain Value Engineering
- 4.1.8 Describe the application of Value Engineering by Software

CONTENT DETAILS

MODULE-I

INTRODUCTION TO QUANTITY SURVEYING

Historical development of quantity surveying. Functions performed by the quantity surveyor in relation to construction works- Evolution of standard methods of measurement for construction Works. The use of document in practice. Basic concepts of quantity surveying. Measurement and computation of lengths, girths, areas, and volumes both for regular and irregular shapes from drawings. Theoretical processes of building contract from inception to completion and the interrelationship of the professional team Engineering drawings for buildings and various structures, Principles of standard methods of measurement of work. Bills of quantities, Drawings. Purpose of Bills of Quantities. Processes of preparing Bills of Quantities, including taking off, working up, Abstracting and billing. Types of bill formats and their uses. Setting out of descriptive and quantitative information in taking off. Dimension, abstracting and Billing Sheets

MODULE-II

Analysis of prices of Construction materials, Labour, Machinery, Overheads & incidentals, Other incidentals in the analysis of rates for various trade items of civil engineering construction, Standard schedule of rates, Rates based on market analysis and their comparison, costing quoting lump sum prices and cost planning, Wastages, taxes, direct and indirect charges and profit, Factors affecting analysis of rates, Analysis of rates for non standard new items of works, Types of construction machinery their uses in various construction activities, Output constants, Life, depreciation, cost of owning and operating charges for machinery, Economics of owning or hiring machinery. Estimating, tendering and contract management Tender structure for different types of contracts, Tendering procedure, Specifications, Preparation of pre-qualification documents, Preparation of Tender Documents and invitation to bid, Submission of bids, Tender documents for LCB, NCB and ICB, Procurement of tender documents, Pre-bid conference and site investigation, Bank guarantees and Performance Bonds, Sub contracting & labour contracting system, Letter of Acceptance and Contract

signing. Study of Contract Conditions, Contract administration, Project Diary, Creating Project Record, Planning at head office of Contractor and preparing for execution, Co-ordination with clients, consultants, other contractors and agencies, Site opening and Mobilisation, Selecting and appointing sub contractors on suitable terms, Contractors need to interact with consultants and the clients, Submission of reports, Submission of bills and follow up for payment, Contract payment, Contract close outs, Defects Notification.

MODULE-III

Law of Surveyors- Valuation of land, buildings and machinery The principal types of landed property, the incidents of their tenure, Income derived from them and the outgoing to which they are subjected; Underlying principles governing the relative values of different classes of property; Annual value and capital value, Methods of Valuation, use of valuation tables; Capital and rental valuation of property; Valuation of life interests in property, Valuation for income tax assessment, valuation for mortgage; Laws governing valuation of land & building; Methods of valuation of Construction machinery.

MODULE-IV

Construction Project Management Concepts: Construction Projects – Objectives, Characteristics, life cycle phases; balancing project time, cost, quality and procurement. Project Management – functions, processes, organization. Project managers role, responsibility, competency criteria; main causes of project failure. Project development process. Project management organization, Project planning and control approach. Project Work Breakdown: Classification of levels, Work breakdown methodology, Assessing duration, Costing activity, CPM/PERT Networks and Precedence Network Analysis, Project Work Scheduling. Types of scheduling : Barchart, Network, Line of Balance schedule, Time limited and Resources limited schedule, Schedule hierarchy, Resources forecasting, Manpower planning, Material planning, Material Procurement schedule, Planning and selection of equipment. Cost planning and Scheduling Computer applications in quantity surveying Quantity estimation, Tendering, Billing and Project Management concept; Quick and accurate quantity calculation; Generation of quantities from graphics from AutoCAD drawing; Graphics generation by taking dimensions from existing AutoCAD drawing; Tabulation of generated quantities in the LBD format of measurement books; Detailed rate analysis of each & every item; Consumption of resources for any estimate; Tendering Facility for contractor to quote the tender and Facility to client for analyzing the tenders; Project management - scheduling & monitoring of activities; Billing - Generating and maintaining measurement books & running bills for project; Pricing the Bill Items; Forecasting the BOQ; Reports generation; Create, manage and understand the status of a contract; Creating Engineers project estimation, bid evaluation; Valuations; Contract administration & field inspection processes; Change order management; Document management; Submittals, invoices; Value Engineering

REFERENCE

- 1.Martin Brook, Estimating and tendering for construction work,Butterworth-4th edition
- 2.Trever Holroyd, Principles of Estimating, Thomas Tesford ,2nd edition
- 3.Aliyan ashworth, Practice and Procedure for Quantity Surveyors,Wileyindia